Project Report

Health Geography

Theme: Mapping and spatial analysis of breast cancer screening uptake in Belgium, 2014 and 2017

Written by:

Badou Zaki Njingou Ndouoya

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List of Abbreviations

IMA: Intermutualistic Agency

HH: High-High

HL: High-Low

I: Moran's Index

LH: Low-High

LISA: Local Indicators of Spatial Association

LL: Low-Low

SCR: Coordinate Reference System

GIS: Geographic Information System

SMR: Standardized Mortality Ratio

1. Introduction

Breast cancer is the most common malignant disease and the leading cause of cancer death among women worldwide (Schoofs et al., 2017; Arbyn et al., 2003). It is also the most prevalent cancer and the leading cause of cancer death among women in Europe (Ferlay et al., 2007). Within Europe, Belgium is the country with the highest risk of breast cancer, with more than 10,000 new diagnoses in 2015 and 2016, corresponding to a global standardized rate (WSR) of approximately 105 per 100,000 person-years (Altobelli et al., 2017). It is slightly above the European average for breast cancer mortality, with 19.4/100,000 women/year and a 15-year survival rate of 75% for all stages combined. Breast cancers are responsible for approximately 3% of all-cause mortality among Belgian women (Desreux et al., 2011).

An effective preventive approach to breast cancer is invaluable. Today, the effectiveness of early detection of breast cancer through mammography has been suggested by several studies. These studies have reported a reduction in mortality rates among women aged 50 to 69 years (Nelson et al., 2009). Furthermore, some authors have found that breast cancer-specific mortality could be reduced by approximately 20% in women over 50 invited for screening (Goossens et al., 2014). To achieve a population-based reduction in mortality, a large proportion of the population must be screened. To obtain a 30% reduction in mortality rates, at least 70% of the target population must be screened every 2 years (Schoofs et al., 2017).

Since 2001, there has been a national early detection program for breast cancer in Belgium. The program was established under the impetus of Europe Against Cancer and complies with criteria established by the European Commission. Women aged 50 to 69 are invited by local authorities and their family physicians for a free mammography at accredited mammography units in their neighborhood. Despite the national prevention program, it appears that a significant portion of the female target population is not screened for breast cancer (Schoofs et al., 2017). It is therefore advisable to identify areas with low participation and to understand these geographical disparities in order to implement targeted and priority interventions to increase adherence to the national breast cancer screening program.

It is in this context that this study set itself the objectives of first mapping breast cancer screening uptake at the municipal level in Belgium (2014 and 2017) using a Geographic Information System (GIS). Then, to identify clusters of municipalities with a significantly lower prevalence of participation in breast cancer screening and to explore explanatory factors through spatial analysis.

2. Methodology

2.1. Type of Study

This is an ecological study based on aggregated data to study the spatio-temporal variations in adherence to breast cancer screening at the municipal level throughout Belgium.

2.2. Data Sources

Breast cancer screening uptake data were extracted from the Intermutualistic Agency (AIM) atlas for 2014 and 2017. These data were collected through the organized national breast cancer screening program. This program is under the responsibility of the different regions.

2.3. Study Population

The eligible population for this study was women aged 50 to 69 who had at least one screening mammography during the year 2014 or 2017 as part of the organized national breast cancer screening program.

2.4. Study Variables

The variable of interest is the adherence rate to the national breast cancer screening program at the municipal level. The explanatory variable used in this study was the average number of patients attributed per active general practice.

2.5. Analysis Plan

2.5.1. Preparatory Analysis

To make our 2014 and 2017 data comparable, indirect standardization was performed to obtain crude prevalences, age-adjusted prevalences, and SMRs (Standardized Morbidity Ratios) of screening participation by municipality. STATA 17 software was used to perform this analysis. A database was created in Excel format containing the different prevalences, the SMR for both years, as well as the difference in adjusted prevalence between 2017 and 2014.

2.5.2. Geocartography

To address our first objective, geocartography using QGIS 3.28 was applied to represent (visualize) crude prevalences, age-adjusted prevalences, and SMRs of adherence to breast cancer screening by municipality in Belgium in 2017. In the presence of quantitative relative data, we produced choropleth maps with discretized value ranges. The distribution of the studied variables was multimodal; therefore, the Jenks discretization method was used to divide the continuous variables into classes. A cool color (blue) was used to represent breast cancer screening adherence, since we are dealing with a prevention method. Lighter colors represented lower prevalences, while darker colors represented higher prevalences. A proportional symbol map (in the form of circles) was created to visualize the average number of patients attributed per active general practice by municipality in 2017. To improve map readability, the average number of patients was aggregated at the district level.

2.5.3. Spatial Analysis

The shapefile of municipalities from the merger was exported to GeoDa 1.20 software to perform spatial analysis of breast cancer screening uptake in Belgium in 2014 and 2017. The Queen contiguity-based weight matrix was generated for calculating spatial autocorrelation statistics. Global spatial autocorrelation was measured through the calculation of the global Moran's I index. Moran's I was permuted 999 times to determine statistical significance, and the threshold was $P < 0.05$. To visualize spatial clusters, Local Indicators of Spatial Association (LISA) were created using local Moran's I calculations. Spatial cluster types High-High (HH) and Low-Low (LL) indicate similar values, and High-Low (HL) and Low-High (LH) indicate dissimilar values. A weighted standard linear regression allowed the selection of the most appropriate spatial model for the analysis. Consequently, the spatial autoregressive model was used to evaluate the spatial association between breast cancer screening uptake and the average number of patients attributed per active general practice in 2017. The significance threshold of 5% was retained for the analysis.

3. Results

3.1. Cartography of the Geographic Variation of Breast Cancer Screening in Belgium

3.1.1. Create a professional map of the municipalities of Belgium clearly distinguishing the different regions

The administrative map of municipalities by region of Belgium is presented in Figure 1 (Appendix 1). Regional boundaries are represented by black lines, and municipal boundaries by light blue lines.

3.1.2. a) Prepare an Excel file containing the project variables, b) Import the municipal basemap and the Excel file into QGIS, c) Join the two files.

The Excel file and the shapefile resulting from the join have been uploaded as an appendix.

3.1.3. Create a map of breast cancer screening participation at the municipal level

a) Crude prevalence of screening participation in 2017

Figures 2 (Appendix 1) present the crude prevalences of breast cancer screening participation in Belgium in 2017. It emerges from this figure that the participation rate is higher in the northeast and northwest of Belgium, specifically in the Flemish region. It is lower in the center and south of Belgium.

b) Standardized prevalence of screening participation in 2017

Figures 3 (Appendix 1) present the standardized prevalences of breast cancer screening participation in Belgium in 2017. The same trends in prevalence distribution are observed as those seen in the crude prevalence figure. Participation rates are higher in the northeast and northwest of Belgium, and lower in the center and south.

c) SMR (Standardized Morbidity Ratio) of screening participation in 2017

The participation rate in almost all of northern Belgium is higher than the national rate for both years, and a lower rate compared to the average is observed in the south and center of Belgium (Figure 4 – Appendix 1).

d) Difference in standardized prevalence between the 2 years (2017 - 2014)

A more markedly negative difference in standardized prevalence is observed in municipalities located in the south and southwest of Belgium between 2014 and 2017 (Figure 5 -Appendix 1).

3.1.4. Create a municipal-level map that represents both screening participation in 2017 (standardized prevalence) and the average number of patients attributed per active general practice

Figure 6 (Appendix 1) presents standardized prevalences of breast cancer screening uptake and the average number of patients attributed per active general practice by municipality, Belgium, 2017. Because the map was very saturated and not interpretable, we represented the number of patients at the district level (Figure 7 - Appendix 1).

3.1.5. Based on the maps created, suggest answers to the following questions:

a) Are there spatial patterns in screening participation and the average number of patients attributed per active general practice?

When analyzing only the distribution of screening uptake prevalences, a spatial pattern is observed with very high participation rates in certain geographic areas and very low rates in others. However, when combined with the average number of patients attributed per active general practice, the spatial structure is no longer observable because some districts have low participation rates but high patient numbers, and vice versa.

b) Is it possible to link the distribution of screening participation and the average number of patients attributed per active general practice?

Based on the map, it seems unlikely because no clear trend is observed that allows linking screening uptake and the number of patients attributed per active general practice.

3.2. Spatial Analysis of Breast Cancer Screening

3.2.1. For the 2 years separately, are there municipalities with "abnormally" high or low values? Which municipalities present these "abnormal" values?

For both years, municipalities with "abnormally" high or low SMR values are observed. In 2014, the municipality of Beaumont had an SMR of 0.076, while the municipality of Ham-sur-Heure-Nalinnes had 2.177. In 2017, the municipality of Bertogne had an SMR of 0, while Ham-sur-Heure-Nalinnes had 2.20.

3.2.2. For the 2 years separately and for the difference, is there a spatial phenomenon in the data?

a. Global spatial autocorrelation

The global Moran's I index for screening data from 2014 ($I=0.857$; $P<0.001$), 2017 ($I=0.854$; $p<0.001$), as well as the difference between these two years ($I=0.395$; $p<0.001$), shows the presence of positive global spatial autocorrelation in screening participation for each of these years (Table 1 - Appendix 2).

b. Local spatial autocorrelation

Figures 8 and 10 (Appendix 2) present Local Indicators of Spatial Association (LISA) for breast cancer screening uptake in Belgium in 2014 and 2017. These figures highlight significant clusters suspected in the Moran scatterplot. For both years, clusters with high screening participation rates are observed in municipalities located in the north and northwest of Belgium. However, clusters with low screening participation rates are observed in municipalities located in the south, southwest, and center of Belgium. For the difference, several clusters with high participation rates are located on a band extending from the northwest to the northeast of Belgium.

Additional information provided by the significance map indicated that 144, 179, and 90 municipalities presented statistically significant local spatial autocorrelation at significance levels of 5%, 1%, and 0.1%, respectively for the year 2014 (Figure 9). For the year 2017, the significance map indicates that 152, 173, and 98 municipalities presented local spatial autocorrelation at significance levels of 5%, 1%, and 0.1%, respectively (Figure 11).

3.2.3. For the most recent year, is there a spatial association between screening uptake and the average number of patients attributed per active general practice?

Ordinary Least Squares (OLS) estimation for spatial dependence diagnostics shows the presence of correlated residuals through a significant residual Moran's I ($I=0.837$; $p<0.001$), indicating the presence of spatial dependence. The spatial regression model was selected based on the Lagrange multiplier test using the decision tree (Table 2 - Appendix 2). The results of the spatial autoregressive model show that screening uptake is significantly associated with the number of patients attributed per active general practice ($\beta=0.002$, $p=0.025$) including a spatial dimension ($Rho=0.905$; $p<0.001$). The likelihood ratio test is significant ($p<0.001$), meaning that the spatial autoregressive model is better than the standard linear model. The introduction of the spatial component in the model explains 84.5% of the variability observed in the model (Table 3 - Appendix 2).

4. Conclusion

At the end of this work, the objectives were, on the one hand, to map breast cancer screening uptake at the municipal level in Belgium in 2017, as well as the difference between 2017 and 2014. On the other hand, to identify clusters of municipalities with a significantly lower prevalence of participation in breast cancer screening and to explore explanatory factors. It appears that breast cancer screening uptake in Belgium is spatially distributed. It is observed that in municipalities located in the north of Belgium (Flanders), participation rates are higher compared to municipalities in the central and southern parts (Brussels and Wallonia), which have lower rates. This geographical disparity could be explained by the fact that screening is a regional responsibility and the strategies implemented differ by region. In Flanders, half of women aged 50 to 69 have had a mammography as part of the organized program. However, in Brussels and Wallonia, most women are screened outside the organized program during consultations with their general practitioner or gynecologist. The spatial association observed between breast cancer screening uptake and the average number of patients per active general practice allows us to say that we can rely on general practitioners to increase screening participation rates. It is therefore necessary to implement a screening policy that includes general practitioners more extensively.

References

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Schoofs J, Krijger K, Vandevoorde J, Devroey D. Health-related Factors Associated with Adherence to Breast Cancer Screening. J Midlife Health. 2017 Apr-Jun;8(2):63-69. doi: 10.4103/jmh.JMH_71_15.

Appendix

Appendix 1: Cartography of the geographic variation of breast cancer screening in Belgium

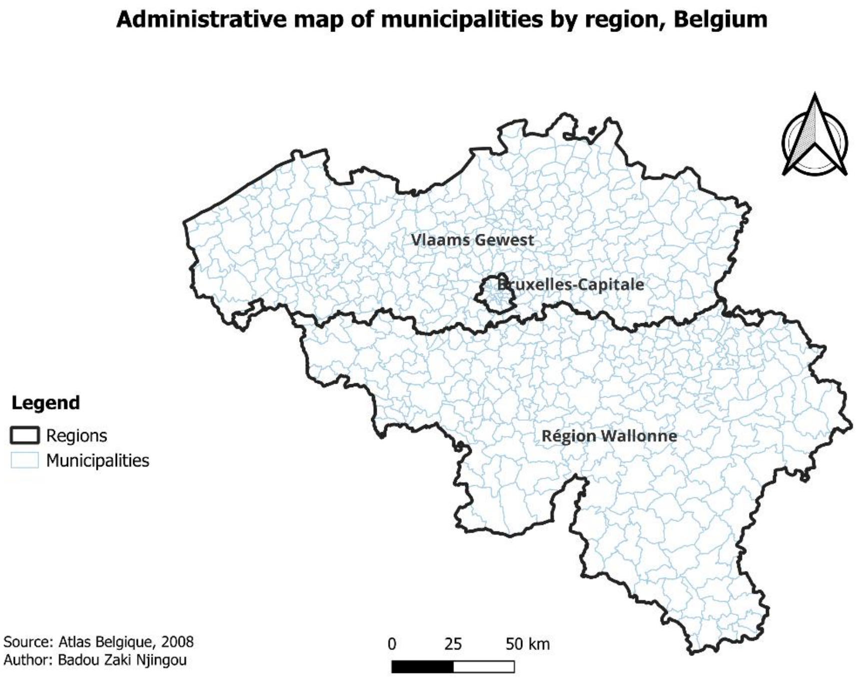

Figure 1: Administrative map of municipalities by region in Belgium

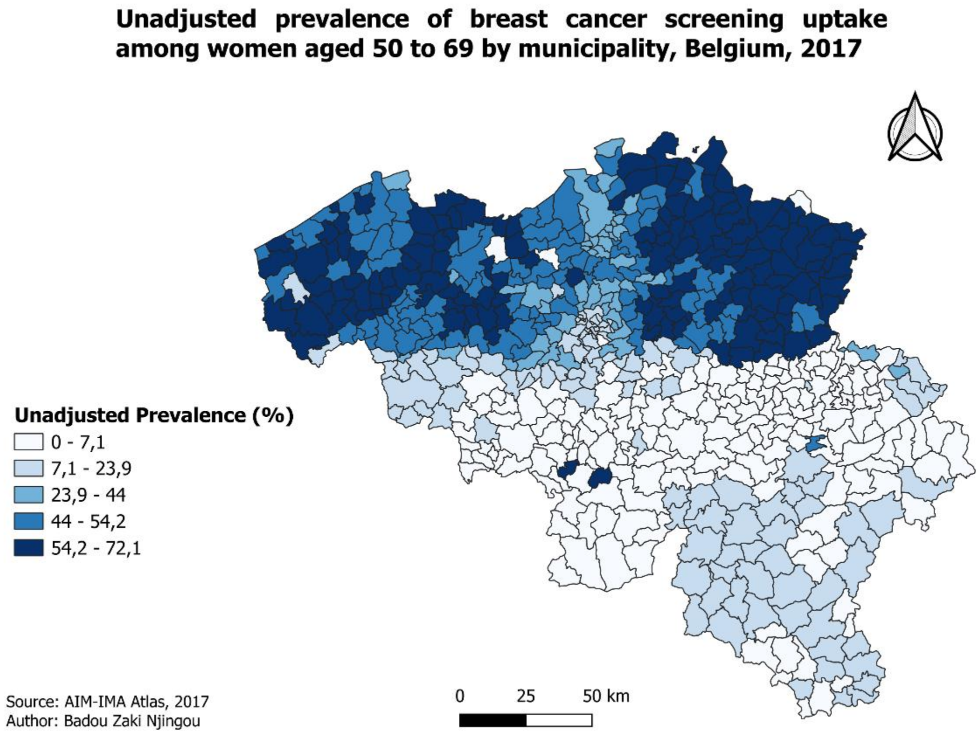

Figure 2: Crude prevalence of breast cancer screening uptake among women aged 50 to 69 by municipality, Belgium, 2017

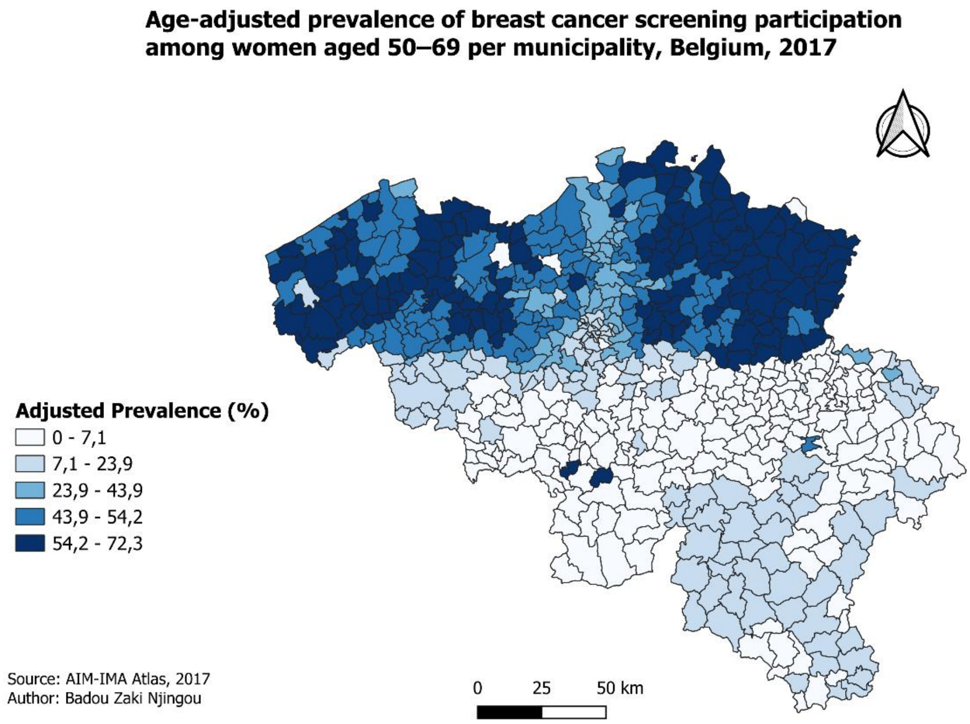

Figure 3: Adjusted prevalence of breast cancer screening uptake among women aged 50 to 69 by municipality, Belgium, 2017

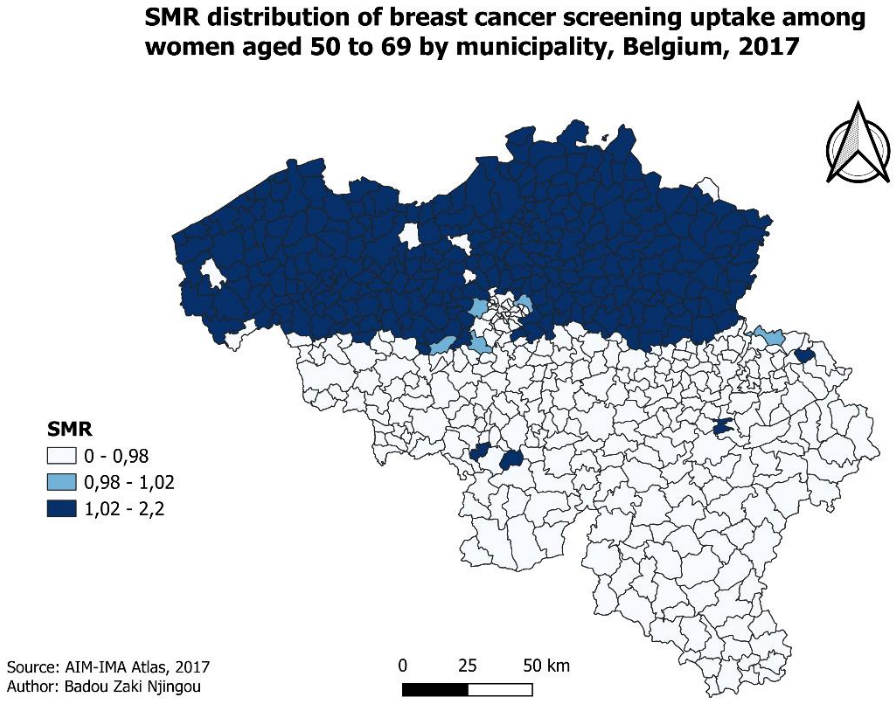

Figure 4: SMR distribution of breast cancer screening uptake among women aged 50 to 69 by municipality, Belgium, 2017

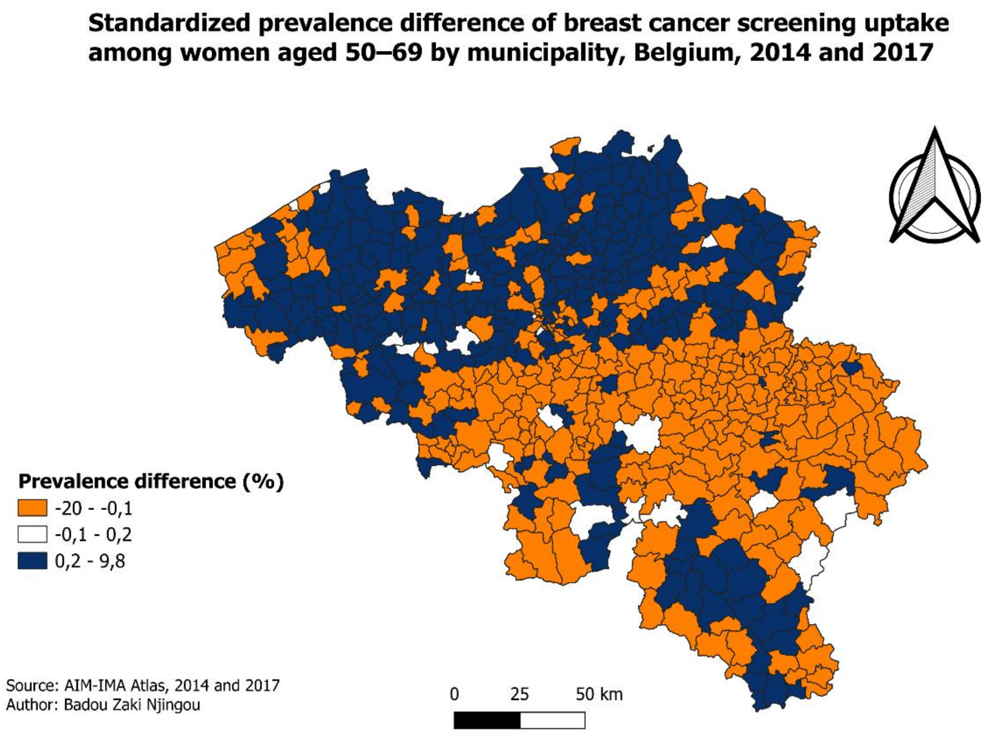

Figure 5: Standardized prevalence difference of breast cancer screening uptake among women aged 50–69 by municipality, Belgium, 2014 and 2017

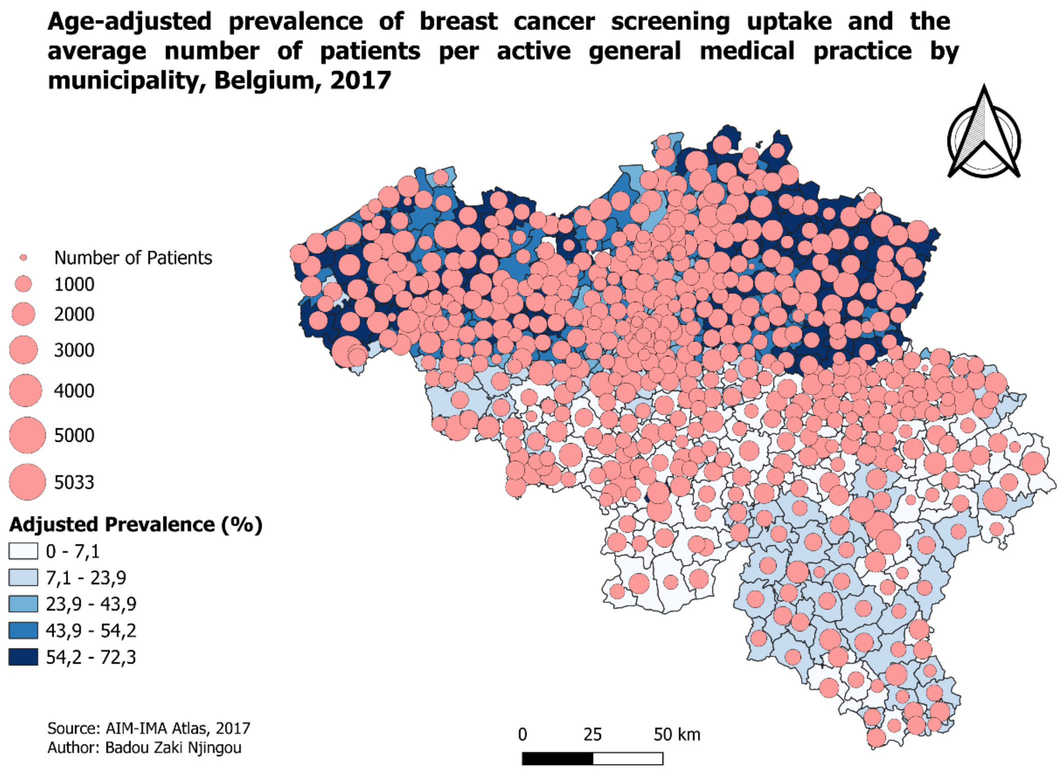

Figure 6: Standardized prevalence of breast cancer screening uptake and the average number of patients attributed per active general practice by municipality, Belgium, 2017

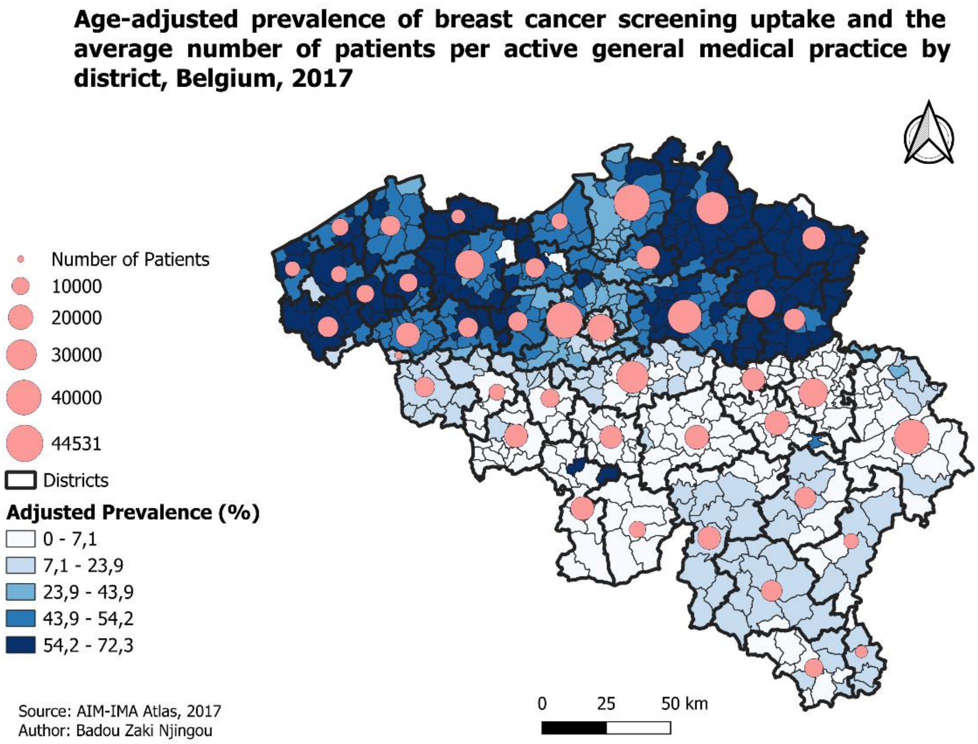

Figure 7: Standardized prevalence of breast cancer screening uptake and the average number of patients attributed per active general practice by district, Belgium, 2017

Appendix 2: Spatial Analysis

Table 1: Global spatial autocorrelation of breast cancer screening uptake in Belgium, 2014 and 2017

Variables	I of Moran	Z-value	P-value
Breast cancer screening uptake in 2014	0.856	34.615	0.001
Breast cancer screening uptake in 2017	0.854	34.506	0.001
Difference in breast cancer screening uptake between 2017 and 2014	0.395	16.094	0.001

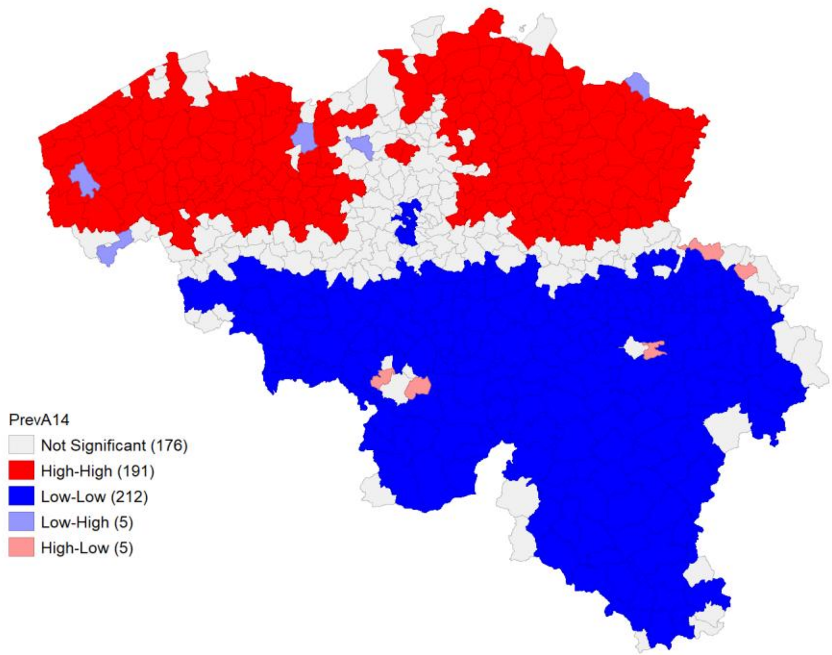

Figure 8: Local indicator of spatial association (LISA) for breast cancer screening uptake in Belgium, 2014 (standardized prevalence)

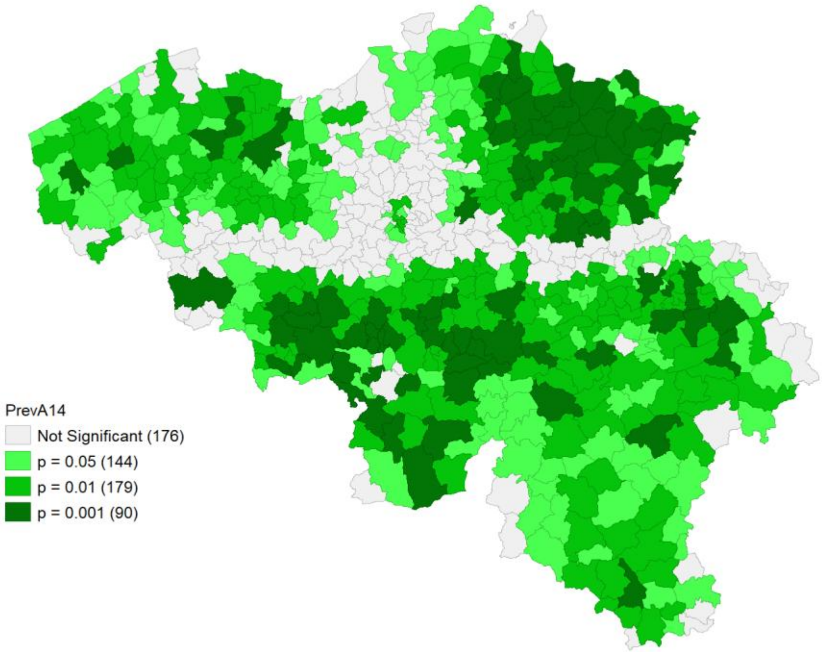

Figure 9: Map of significant clusters of breast cancer screening uptake in Belgium, 2014 (standardized prevalence)

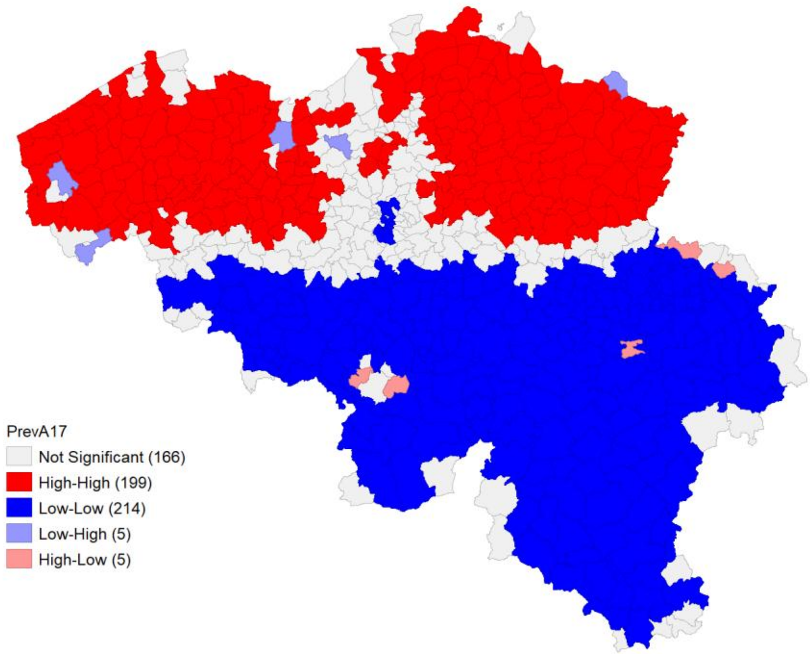

Figure 10: Local indicator of spatial association (LISA) for breast cancer screening uptake in Belgium, 2017 (standardized prevalence)

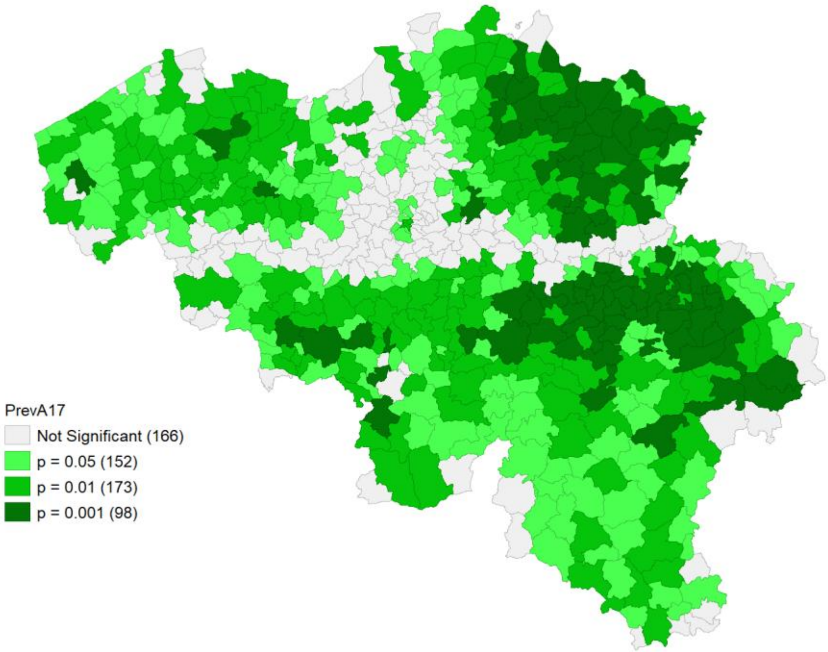

Figure 11: Map of significant clusters of breast cancer screening uptake in Belgium, 2017 (standardized prevalence)

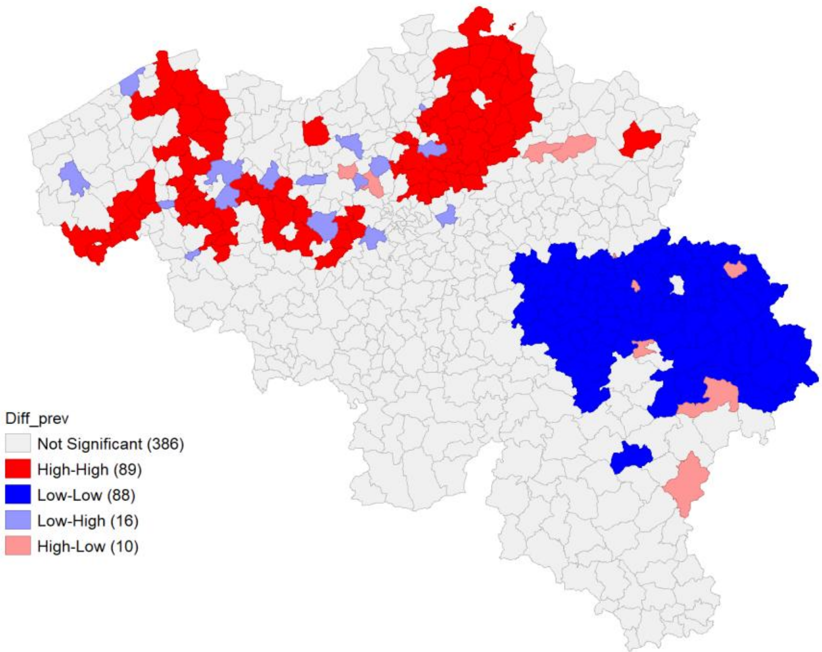

Figure 12: Local indicator of spatial association (LISA) for breast cancer screening uptake in Belgium, 2017-2014 (standardized prevalence difference)

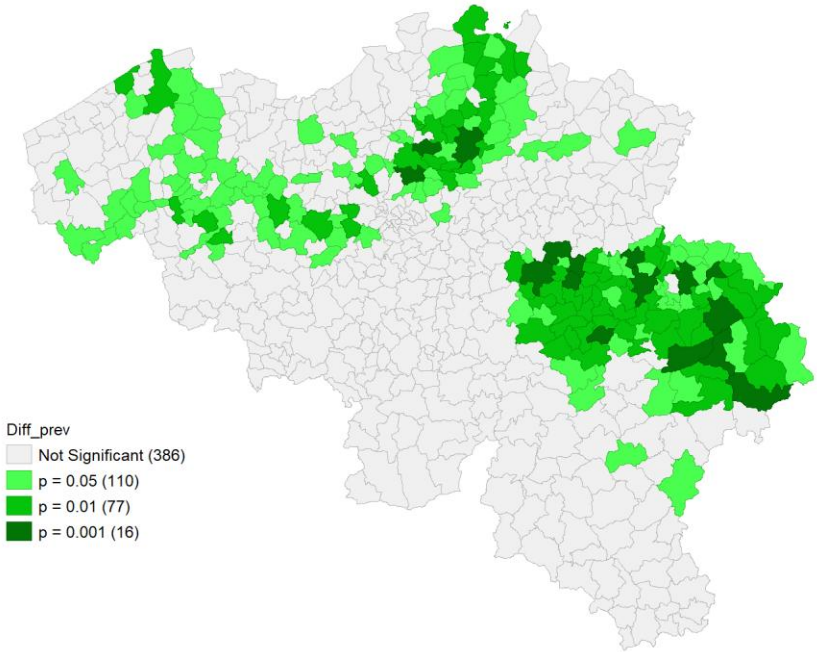

Figure 13: Map of significant clusters of breast cancer screening uptake in Belgium, 2017-2014 (standardized prevalence difference)

Table 2: Standard linear regression model for diagnostic testing of spatial dependence

Variables	Coefficient	Standard Error	P-value
Number of patients	0.007	0.002	<0.001
Constant	20.325	2.856	<0.001
Spatial Dependence Diagnostics			
Residual Moran's I	0.837		<0.001
Lagrange Multiplier Test (lag)	1122.45		<0.001
Lagrange Multiplier Test (error)	1090.22		<0.001
Robust Lagrange Multiplier Test (lag)	37.97		<0.001
Robust Lagrange Multiplier Test (error)	5.74		0.017
Fit Measures			
R^2	0.02		
AIC	5368.24		
BIC	5376.99		

Note: AIC: Akaike Information Criteria; BIC: Bayesian Information

Table 3: Association between screening uptake and the average number of patients attributed per general practice in Belgium, 2017

Variables	Coefficient	Standard Error	P-value
Number of patients	0.002	0.001	0.025
Constant	0.346	1.237	0.780
Rho	0.905	0.179	<0.001
Fit Measures			
LR test	935.58		<0.001
AIC	4434.66		
BIC	4447.78		
R^2	0.845		

Note: LR: likelihood ratio; AIC: Akaike Information Criteria; BIC: Bayesian Information Criteria